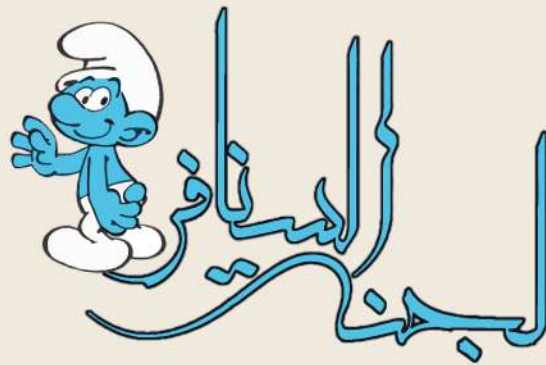


أسئلة سنوات فاينال تقنيات عددية



Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20

Q1: When doing two bisection iterations to the function $f(x) = x^2 - 2$ on the interval $[1, 2]$, we get

- (A) 1.75 (B) 1.25 (C) 1.5 (D) none

Q2: For some differentiable real function f , assume that

$$\frac{f'(x)}{f(x)} = \frac{e^x + e^{-x} + 2}{e^x - e^{-x} + 2x}$$

Using the initial value $x_0 = 1$, the first Newton-Raphson iteration leads to

- (A) 0.8184 (B) -0.1447 (C) 0.1447 (D) none

Q3: The system $\begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ has

- (A) no solution (B) two solutions (C) infinitely many solutions (D) unique solution

Q4: The Cholesky decomposition cannot be applied on the matrix

- (A) $\begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & -3 \\ -3 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$

Q5: For some real observations $\{(x_i, y_i)\}_{i=1}^n$ when we use the least squares technique on the curve fitting $y = ax^3$; then a equals

- (A) $\frac{\sum y_i x_i^3}{\sum x_i^6}$ (B) $\frac{\sum y_i x_i^4}{\sum x_i^9}$ (C) $\frac{\sum y_i x_i^3}{\sum x_i^7}$ (D) none

Q6: If we need to approximate a function f by an n -th order interpolation polynomial " $f_n(x)$ ", which of the following is wrong

- (A) Higher degree of the interpolation polynomial improves the accuracy
- (B) To find $f_n(x)$ we need more than $n + 1$ different points that are located on the function f
- (C) Both Newton divided difference and Lagrange methods yield the same interpolation polynomial $f_n(x)$
- (D) Assume that we used Lagrange method to derive $f_n(x)$. If we want to add a new point that is located on the function f , then we cannot use the computations of $f_n(x)$ to derive $f_{n+1}(x)$

Q7: Using the Simpson's $\frac{3}{8}$ rule to approximate $\int_0^3 \sin(x) dx$ gives

- (A) 4.45
- (B) 6.068
- (C) 2.023
- (D) none

Q8: Suppose that we want to approximate $\int_a^b f(x) dx$ using 10 subintervals composite (multiple application) Simpson's $\frac{1}{3}$ rule. To this end; we replace; in each subinterval, f by a polynomial of the degree

- (A) 1
- (B) 2
- (C) 10
- (D) 20

Q9: The slope of the secant line through the points (a, a) and $(a + 1, c)$ equals 1. Assume that the aforementioned two points are located on some real function f and that the trapezoidal rule for approximating $\int_a^{a+1} f(x) dx$ gives 2. Then the value of the constant a equals

- (A) 1.5
- (B) -0.5
- (C) 0.5
- (D) -1.5

Q10: For some constant c and differentiable function f , assume that

$$f(c - 0.1) + 1 = f(c) = f(c + 0.1) + 2$$

Then, the forward difference approximation of $f'(c)$ equals

- (A) -5
- (B) 10
- (C) -20
- (D) none

Q11: For the I.V.P: $y' + 2y = 0$; $y(0) = 1$, let y_1 be the Euler's approximation of $y(0.1)$ when the step size $h = 0.1$ is used. Then $y_1 =$

- (A) 0.8
- (B) 1.6
- (C) 2.4
- (D) none

Q12: For the I.V.P: $y' + 2y = 0$; $y(0) = 3$, let y_i be the Heun's $\alpha_2 = 0.5$ special case of the 2nd order Runge-Kutta prediction of $y(0.2)$ when the step size $h = 0.2$ is used. Then $y_1 =$

- (A) 2.64 (B) 2.04 (C) 2.84 (D) none

Q13: For the B.V.P $\frac{d^2 T}{dx^2} + \lambda(T_a - T) = 0$, $T(0) = 0$, $T(3) = a$. Let $\lambda = T_a = 1$. By using the Finite-Difference Approximation

$$-T_{i-1} + (2 + \lambda(\Delta x)^2) T_i - T_{i+1} = (\Delta x)^2 T_a \lambda, \quad i = 1, 2, \dots$$

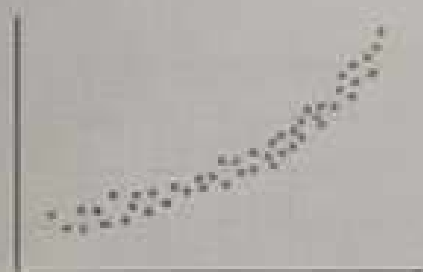
on two interior nodes we get $T(2) = \frac{2}{3}$. Then the value of the constant a equals

- (A) $\frac{2}{3}$ (B) 2 (C) 1 (D) none

Q14: Let f be some real polynomial. If the trapezoidal rule yields exact result when approximating $\int_a^b f(x) dx$, then the degree of the polynomial f is

- (A) 0 but not 1 (B) 1 but not 0 (C) 0 or 1 (D) none

Q15: The data points $\{(x_i, y_i)\}_{i=1}^n$ are plotted in the following figure



The most appropriate curve fitting equation for the set of data is

- (A) Exponential (B) Logarithmic (C) Linear (D) Trigonometric

Q16: If we apply the simple fixed point method and initial value $x_0 = 1$ on the equation $x = g(x) = \frac{3}{x}$, then $x_{100} =$

- (A) 3 (B) 1 (C) 2 (D) none

Q17: The 3 digits rounding approximation of e is

- (A) 2.72 (B) 2.71 (C) 2.718 (D) none

Q18: When doing ONE Gauss-Seidel iteration on the linear system

$$2x = y + z, \quad y = 2x + z - 3, \quad z = 3x - 2y$$

using the initial values $y_0 = z_0 = 0$, we get $z_1 =$

- (A) 2 (B) 6 (C) -7 (D) none

Q19: For some 100×100 tridiagonal matrix $[A]$, which of the following is false

- (A) The number of Thomas-Algorithm iterations is 100
(B) The Thomas Algorithm does not affect the entry in the first row and first column of $[A]$
(C) The Thomas Algorithm does not affect the upper diagonal of $[A]$
(D) The Thomas Algorithm can be used to derive the LU -decomposition

Q20: Let f be a third order polynomial that passes through the points $(1,0)$, $(3,0)$ and $(5,0)$, then which of the following is true

- (A) $f(0)f(7) < 0$
(B) $f(0)f(7) > 0$
(C) $f(0)f(2) > 0$
(D) $f(0)f(4) < 0$

GOOD LUCK



Al-Balqa' Applied University, Faculty of Engineering Technology
Final Exam, Numerical Techniques

Student Name:

Instructor name:

Lecture Time:

Date: 4 Jan. 2017

Question One : For parts (i)-(iii), put a circle around the correct answer

(i) When approximating the definite integral $\int_a^b f(x)dx$ by trapezoidal rule ; one replaces the function $f(x)$ by

- (a) $f_1(x)$ (b) $f_2(x)$ (c) $f_3(x)$ (d) $f_4(x)$ (e) none

(ii) When doing one fixed point iteration to approximate a solution of the equation $x = \cos(x)$ using the initial value $x_0 = 1$, we get $x_1 =$

- (a) 0.5403 (b) -0.1585 (c) -0.4597 (d) 0.8415 (e) none

(iii) For the polynomial $f(x) = 3x^5 + 2x + 2x^4 + 1$, when we use the fifth order Taylor series expansion to approximate $f(1)$, the absolute error becomes

- (a) 5 (b) 1 (c) 0 (d) -1 (e) none

Question Two : For some infinitely many differentiable function f , assume that $f(1) = 2$ and $f(0.8) = 1$. Use backward difference approximation to estimate $f'(1)$

Question Three : Consider the I.V.P: $y' = xy$; $y(0) = 2$ and the step size $h = 2$.

(i) Use Euler's method to predict $y(2)$.

(ii) Write the formulas of the Midpoint Method $a_2 = 1$ that is a special case of the second-order Runge-Kutta methods.

(iii) Use the answer of (ii) to predict $y(2)$

Question Four: Approximate $\int_0^2 e^{x^2} dx$ using Simpson's $\frac{1}{3}$ rule

Question Five: Consider the B.V.P

$$\frac{d^2 T}{dx^2} + h'(T_a - T) = 0$$

$$T(0) = 0, \quad T(10) = 2$$

Use the Finite-Difference Approximation of Boundary-Value Problems:

$$-T_{i-1} + (2 + h'(\Delta x)^2) T_i - T_{i+1} = (\Delta x)^2 T_a h', \quad i = 1, 2, \dots$$

to approximate $T(5)$ knowing that $\Delta x = 5$, $h' = 1$ and $T_a = 1$

Question Six: For the system

$$x = y + z$$

$$y = 2z - x + 1$$

$$z = (x + y)/3$$

Do one Gauss-Seidel iteration to find z_1 using the initial values $y_0 = 1$ and $z_0 = 0$

Question Seven: Approximate $\int_0^{2\sqrt{a}} \sin(x^2) dx$ using two segments composite trapezoidal rule! (the answer has to be in terms of the positive constant a)